Pagemark - Roadmap (rewritten 2026-10-14, after v0.9.0)

Written to be picked up cold, from another computer or by a fresh Claude Code session. Pair it with CLAUDE.md (architecture and every gotcha) and ROUNDING_OUT_REPORT.md (the 430-item backlog; IDs like `FB-24` below refer to it).

Direction (decided with the owner): Pagemark is for **non-technical readers**. Plain words everywhere, nothing that needs a developer account or a setup guide, private by default, no paid services, no cloud account required.

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1. Shakedown: where the app stands

What is solid

Area	State
Core log	Books, reads (print / e-book / audio), stars, thoughts, progress, goals, series, shelves, quotes, author pages, detailed ratings, 5 library styles, 6 themes, trash, undo
Privacy	Locked library (PIN or fingerprint, recovery code, lockouts), optional whole-app lock, no screenshots/recents of locked screens, hidden Locked tab, widget never shows locked books
Data safety	Verified restore with restore points, password-protected backup file, spreadsheet export, import from Goodreads/StoryGraph/Pagemark with undo, schema migrations 1 to 5 tested
Quality	228 automated tests, analyzer clean, 84-finding audit (all HIGH and MEDIUM fixed in code), 5,000-book library tested (cold start 1.3 s)
Product	Settings in plain language, built-in guide kept honest by a test, notifications page, share target, widget, shortcuts
Release process	`tool/build.ps1` makes signed APKs and a Play bundle; 9 releases since the audit

What is weak or unproven (honest list)

1. **Nothing has had a real-phone test pass.** Fingerprint, the barcode scanner, the widget, the 10:00 reminders, the in-app update install and restore on a second phone were only checked on an emulator, or not at all (`docs/release-audit/MANUAL_TEST_PLAN.md`).
2. **Backups depend on the user.** Google Drive backup was removed (too hard to set up). The reminder helps, but a person who ignores it has no safety net, and "save a file somewhere" is still a technical idea.
3. **First run is bare.** An empty Library and a button. No welcome, no sample data, no explanation of the Locked tab.
4. **Finding a book can fail.** One search source (Open Library). Obscure titles, new releases and non-English books often come back empty, which a non-technical person reads as "the app is broken".
5. **Accessibility is unchecked.** No TalkBack pass, large text only partly checked, tap sizes unmeasured. Many readers are older or low-vision.
6. **English only, phone portrait only.** No localisation, no tablet or landscape layouts.
7. **Documentation is stale.** Manual PDFs and screenshots are from 0.4 to 0.7; the UI layout guide predates the new Settings, Notifications and ratings.

8. **Code health.** `lib/ui.dart` is about 1,700 lines holding most screens. Hard to change safely. A script silently garbled dashes in 0.7.1 to 0.8.0 and no test caught it.
9. **Audit leftovers.** 22 LOW and 12 INFO items are accepted or deferred (see `docs/release-audit/FINDINGS.md`).
10. **Not on any store.** Play Store is prepared (`docs/PLAY_STORE.md`) but nothing is registered. iOS needs a Mac.

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2. Proposed order

0.9.1 "Prove it" (small, do first)

Goal: find out what is really broken before adding more. - **Real-phone test pass** on your own phone using `docs/release-audit/MANUAL_TEST_PLAN.md`: fingerprint, scanner, widget, reminders arriving, update install, restore on a second phone, TalkBack basics. Record results in the repo. - Fix whatever it finds. - **Mojibake guard:** a test that fails if source files contain garbled dashes or a BOM (the 0.7.1 bug). - Widget preview image in the widget picker. - Regenerate screenshots and the UI layout PDF for the current screens.

0.10.0 "First run and safety" (the non-technical release)

- **Welcome screens** (3 swipes, skippable) and an **optional sample library** ("try it with 5 example books", removable in one tap). Explains the Locked tab and backups in one line each.
- **Backup to a folder you choose (FB-38).** Pick a folder once with Android's own file picker; Pagemark then saves its password-protected backup there by itself (when you open the app and a backup is due). Because Android's picker can point at **Google Drive, OneDrive, Dropbox or a USB drive**, this gives cloud backup with **no Google Cloud project, no client ID and no sign-in code in Pagemark**. This replaces the removed Drive feature. Needs a small Android file-access piece (about one session).
- **"Back up now?" nudge** in the Library when overdue, not just a notification.
- **Smarter search when nothing is found:** offer to try another spelling, search by author, or add by hand with the barcode number kept.
- **Defaults (FB-88):** date format, week start, default format (print/e-book/audio).

0.11.0 "Everyone can use it"

- **Accessibility pass (FB-86):** TalkBack labels for every control, 48 dp tap targets, tested at 200 % text, contrast check, no information carried by colour alone.
- **Tablet and landscape layouts (FB-84):** two-pane Library and book page at 840 dp and up.
- **Localisation (FB-85):** move all text to translation files; English plus one more language (owner to choose); check right-to-left. The in-app guide is translated too.
- **Code health:** split `ui.dart` into one file per screen group, no behaviour change, tests unchanged.
- Close out the deferred audit items that are cheap.

0.12.0 "Reading life" (features people enjoy)

Chosen for non-technical readers: visible, friendly, no setup. - **Year in books (FB-34 / FB-58):** a shareable picture card ("2026: 24 books, 8,120 pages, favourite genre Fantasy"), spoiler-free and never including locked books. - **Reading streak and calendar (FB-09):** a gentle month view of days you read, with no guilt wording. - **Reading timer (FB-02):** start/stop while you read, pages per session feeding the progress card. - **To-read list with order (FB-10):** drag to reorder what to read next. - **Lending tracker (FB-67 / FB-68):** "lent to Sam on 3 Oct", with an optional reminder to ask for it back. - **Printable reading report (FB-33):** one PDF for the year.

0.13.0 "Finding books"

- **Second book source (FB-15):** Google Books alongside Open Library, merged, so fewer searches come back empty.
- **Refresh details (FB-17)** for an existing book without losing your data; **edition picker (FB-16)**.
- **Bulk scan mode (FB-18):** scan a whole shelf, one beep per book.
- **"What next?" suggestions (FB-24 / FB-25)** from your own library only (same author, series, genre); nothing leaves the phone.

1.0.0 "Done"

Exit criteria: everything above shipped or consciously dropped; no open data-loss or leak finding; a clean real-phone pass; screenshots cover every screen. - **Google Play release:** developer account, a 14-day closed test with about 12 testers, Data safety form (answers prepared), listing and graphics. See `docs/PLAY_STORE.md`. - **Major-release documentation set (house rule):** regenerate the User Manual, Install Manual, Release Notes and README; write the **Package Overview in three tiers** (simple, medium, advanced); refresh the GitHub repo details (description, topics, screenshots). `python docs/_tools/build_docs.py`, `docs/_tools/build_ui_layout.py`.

After 1.0

- **iOS (FB-83):** needs a Mac (Face ID text, Keychain, share extension, TestFlight).
- Chapter notes with spoiler hiding (FB-07), merge-restore (FB-41), edition-level dedupe (FB-23), price tracking (FB-66).

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3. What I would not do (and why)

Idea	Decision
Encrypted database (SQLCipher)	Declined by the owner, 2026-10-12. Do not re-propose.
Google Drive API backup	Removed in 0.9.0: needs a Cloud project and client ID. Replaced by "backup to a folder you choose" (0.10.0).
Accounts, social features, ads, analytics	Against the product's promise.
Decoy PIN, multiple vaults, custom fields, badges and challenges, LAN sync between phones	Too technical or too niche for the audience; parked.
Paid services or certificates	Never (owner rule).

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4. Working away from home

Get a working checkout (any OS)

```
git clone https://github.com/AxialForge/Pagemark.git && cd Pagemark
flutter --version                    # needs Flutter 3.47+ (stable)
flutter pub get
dart run build_runner build --delete-conflicting-outputs   # regenerates lib/db.g.dart
flutter analyze && flutter test      # must be green before you start (about 230 tests)
```

- **Windows + JDK 25 only:** Gradle fails with "Unable to establish loopback connection". `mkdir C:\tmpj` and set `JAVA_TOOL_OPTIONS=-Djdk.net.unixdomain.tmpdir=C:/tmpj` before any `flutter build`.
- **No phone?** Use an Android emulator (API 34+ x86_64). Helpers and the screenshot recipe are in `docs/_tools/screenshots/` and `docs/_tools/build_ui_layout.py`.

What needs the home PC

Anywhere	Home PC only
All code, tests, docs, emulator screenshots, debug APKs, pushing branches	The signing keystore `~/.android/pagemark-release.jks` and `android/key.properties` (never in git) and `tool/build.ps1` release builds
CI runs (analyze, tests, debug builds of both flavors)	Real-phone checks and anything using your accounts

House rules

- Commit as **AxialForge** (check `git config --local --get-regexp "^user\."` is empty). No attribution trailers. Imperative messages, no `feat:` prefixes.
- Bump `pubspec.yaml` and write the `CHANGELOG.md` entry in the same commit; tag `vX.Y.Z`; attach the two APKs (`arm64`, `armv7` must stay in the file names, the updater matches them).
- **Definition of done for any task:** analyzer clean, tests green with new tests; locked data never leaks (grep for places that list books); **update `assets/guide.md`** (the in-app guide, checked by `test/guide_test.dart`) and the changelog; schema change means a migration step plus a test (see the migration trap in CLAUDE.md).
- Never edit source with Windows PowerShell `Get-Content`/`Set-Content` (re-encodes UTF-8).
- Plain words in every message a user can see.

A prompt to paste into Claude Code on another machine

> Read CLAUDE.md and docs/ROADMAP.md. Implement the next unfinished item of release **0.9.1** (or the release I name) on a new branch, with tests, following the house rules. Run `flutter analyze` and `flutter test`. Update `assets/guide.md` and `CHANGELOG.md`. Don't tag or publish; leave the branch pushed and summarise what I should test on my phone.